AKENTEN APPIAH-MENKA UNIVERSITY
of Skills Training *and* Entrepreneurial Development

Hello !!!

Silence Hour

Use this time to revise your notes before we begin.

AKENTEN APPIAH-MENKA UNIVERSITY
of Skills Training *and* Entrepreneurial Development

ICT233

Web Technologies and Design

Lecture Five

Today's Lecture

1.0 Presenting data in Tables using HTML

2.0 Accepting User Inputs with HTML Forms

3.0 Introduction to Scripting

Lecture Objectives

- By the end of the lecture, the student will be able to:
  - Present data in tables on web pages.
  - Design forms for user inputs.
  - Explain and differentiate between server-side and client-side scripting.
  - Design user interfaces that accept inputs from users and display the result to users using scripts.

Presenting Information in Tables

- Tables provide an excellent way to organize and display information on web pages.
- Tables are defined using the **<table> </table>** tag pair.
- A table is divided into rows with the **<tr> </tr>** tag pair, and each row is divided into data cells using the **<td> </td>** tag pair.
- The letters **td** stand for "table data" which is the content of a data cell.
- A data cell can contain text, images, lists, paragraphs, forms, horizontal rules, tables, and so on.

Presenting Information in Tables

- The HTML **<table> element** contains the table information, which consists of **table header elements (<th>)**, **table row elements(<tr>)**, and individual **table data cells(<td>)**.
- These are the three most commonly used HTML elements for presenting data on web pages.
- Apart from these three, you will also see the semantic elements **<thead>** , **<tbody>**, and **<tfoot>** indicating the table header, body and footer information, respectively.

HTML Table Structure

```
<table border= "1" >
    <tr> <!-- Table Row 1 -->
        <td>Cell 1 data</td>
        <td>Cell 2 data</td>
        <td>Cell 3 data</td>
    </tr> <!– End of Table Row 1 -->

    <tr><!-- Table Row 2 -->
        <td>Cell 4 data</td>
        <td>Cell 5 data</td>
        <td>Cell 6 data</td>
    </tr> <!– End of Table Row 2 -->

</table>
```

Cell 1 data	Cell 2 data	Cell 3 data
Cell 4 data	Cell 5 data	Cell 6 data

HTML Table Elements

Element	Description
<table>…..</table>	Establishes the table. It contains all other elements that specify caption, rows, and content
<tr>…..</tr>	Table row. It contains the table cells.
<td>……</td>	Table data cell. This contains the actual table data.
<th>……</th>	Table header cell: It contains header information for a column of data.
<caption>….</caption>	Provides a short description of the table's contents.
<thead>…..<thead>	Signifies table header.
<tbody>…..<tbody>	Signifies table body
<tfoot>……<tfoot>	Signifies table footer

Using the border and width Attributes

- The border attribute is used to specify the thickness of the lines that make the table.
- The width attribute specifies how wide the table should be in the browser window or relative to other layout elements.
- It is a good practice to specify the table width in percentages.
- The example below sets values the border and the width attributes of a table.

```
<table border="1" width="35%">
        Other codes that make up the table go here
</table>
```


HTML Table Example

My Favorite Collections

Title of Music	Artiste
Forward to Africa	Joseph Hill
China Roses	Enya
Floral Street	Enya
Humble Africa	Joseph Hill
Nah Stay Inna Babylon	Joseph Hill
Miss Independent	Neyo
Storm in Africa	Enya
Count on Me	Whitney Houston

```
<!DOCTYPE html>
<html>
<head> <title> HTML Tables: </title> </head>
<body>
    <table  width="55%" border="1" style="border-collapse:collapse;">
    <caption>My Favorite Collections</caption>
      <tr>  <th>Title of Music </th> <th> Artiste </th>  </tr>
      <tr>  <td> Forward to Africa </td>  <td> Joseph Hill </td>  </tr>
        <tr>  <td> China Roses </td>  <td> Enya </td>   </tr>
        <tr>  <td> Floral Street </td>  <td> Enya </td> </tr>
        <tr> <td> Humble Africa </td>  <td> Joseph Hill </td> </tr>
        <tr>    <td> Nah Stay Inna Babylon</td>  <td> Joseph Hill </td> </tr>
        <tr> <td> Miss Independent </td>  <td> Neyo </td> </tr>
        <tr> <td> Storm in Africa</td> <td> Enya </td>  </tr>
        <tr> <td> Count on Me</td> <td> Whitney Houston</td> </tr>
        </table>
    </body>
</html>
```

Spanning Columns

- The **colspan** attribute lets you create cells that span multiple columns of a table.
- Column cells always span to the right.
- The syntax is given as:

```
<tr>
        <td colspan="number of columns"> Content goes here.</td>
</tr>
```

- **Example:**

```
<tr>
        <td colspan="3">
                This text spans across 3 columns.
        </td>
</tr>
```


Spanning Rows

- The **rowspan** attribute lets you create cells that span multiple rows of a table.
- Rows always span down.
- The syntax is given as:

```
<tr>
        <td rowspan="number of rows"> Content goes here.</td>
</tr>
```

- **Example:**

```
<tr>
        <td rowspan="4"> This text spans across 4 rows down.</td>
        <th>Name</th>
        <th>Phone</th>
        <th>Email</th>
</tr>
```

Organizing HTML Table Contents

- For code organization, rows can be grouped into head, body, and footer sections using the **<thead>**, **<tbody>**, and **<tfoot>** elements respectively. E.g

```
<table>
    <thead>
        <tr><th>Header Cell 1</th> <th>Header Cell 2</th> </tr>
    </thead>
    <tbody>
        <tr> <td>Body Cell 1</td> <td>Body Cell 2</td> </tr>
        <tr><td>Body Cell 3</td> <td>Body Cell 4</td> </tr>
    </tbody>
    <tfoot>
        <tr><td>Footer Cell 1</td> <td>Footer Cell 2</td> </tr>
    </tfoot>
</table>
```


Applying Style to the Table

- Styling table elements using CSS enhances the appearance of the table, and the information presented in it.
- The CSS style rules could be specified within the head section of the HTML file using the **<style>** element.
- We provide a CSS style rule code segment that styles any given HTML table on the next slide.

Sample CSS rule

```
<style type="text/css">
    table{
        border: solid 1px black;        border-collapse: collapse;
        font-family: Georgia, Serif, "Times New Roman";  color: #722750;
        }
    td, th {
        border: solid 1px black;        padding:  5px;
        }
    th  {
        font-family: "Trebuchet MS", Georgia, Serif, Verdana;        font-size:  14px;

        font-style:  bold;    background-color:  #7fa2c1;    color:  #0070dd;
        }
    td  {
        font-family: Georgia, Serif, Verdana;    font-size: 12px;
        background-color: #ccdae6;
        }
</style>
```


Creating Background Hover Effects

- Hover effects could be used to add interactivity to tables by using CSS rules.
- When a user hovers the pointer over a cell or row then the background, font, or formatting can change.
- An example is shown below:

```
td  {
        background-color:  #ccdae6;
    }
td:hover  {
        color: white;
        background-color:  #722750;
        }

tr:hover  td {
        color: white;
        background-color:  #722750;
        }
```

Accepting User Input with HTML Form

- An HTML form is part of a web page that includes areas where users can enter or choose information.
- Forms are used by web developers to collect information from users.
- Information submitted by users via HTML forms are processed by servers that host the page by using scripts.
- ***Scripts*** are programs that process user supplied data (via forms) based on the underpinning business logic and send a response to the user via the browser.

The <input /> tag

- The most used form tag is the **<input />** tag.
- The input type expected of the user or the form control (such as text box or button) is specified with the **type** attribute.
- The following are the most commonly used values for the type attribute: **text**, **password**, **submit**, **file**, **hidden** etc.

Common type values for the <input /> tag.

Type Value	Description
text	Defines a single line text box
file	Defines a textbox with a browse button. (For file uploads)
hidden	Defines a hidden text field
checkbox	Defines a checkbox.
radio	Defines a radio button.
submit	Defines a pushbutton.
reset	Defines a reset button.
password	Defines a password field.
Image	Defines an image button
Button	Defines a button

Different HTML Input types - 1

```
<input type="text" />
<input type="password" />
<input type="radio" />
<input type="range" />
<input type="reset" />
<input type="search" />
<input type="submit" />
<input type="tel" />
<input type="time" />
<input type="url" />
<input type="week" />
```

Different HTML Input types - 2

<input type="button" />
<input type="checkbox" />
<input type="color" />
<input type="date" />
<input type="datetime-local" />
<input type="email" />
<input type="file" />
<input type="hidden" />
<input type="image" src="path to image file" />
<input type="month" />
<input type="number" />

Some Common Form Elements

TAG	DESCRIPTION
<form> </form>	Defines a form for user input
<input />	Defines an input field
<textarea> </textarea>	Defines a textarea (a multiline text input control)
<label> </label>	Defines a label to a form control
<fieldset> </fieldset>	Defines a fieldset around a number of form controls
<legend> </legend>	Defines a title for a fieldset
<select> </select>	Defines a selectable list(a drop-down box)
<option> </option>	Defines an option in a drop-down box

Using the <select> Element

- The **<select> ..</select>** element makes it possible to create a list of items from which end users make choices.
- The **<select> </select>** tag pair is used with the **<option> </option>** tag pair to create the list of items.

```
<form  action="display.php" method="post">
    <p>Select your favourite car:</p>
    <select >
        <option>BMW 7 Series </option>
        <option>Haval H9</option>
        <option>Honda CRV </option>
        <option>Lexus GX 470</option>
        <option>Mercedes Benz S550</option>
        <option>Toyota Camry </option>
    </select>
</form>
```


The <textarea> Element.

- The **<textarea>** element lets you create a text area for user inputs larger than a simple text box
- The cols and rows attributes are used to specify the width and height of the text area.
- Using a textarea (a multiline text input control), a user can write an unlimited number of characters.

```
<p>Please tell us  your story here:</p>
<textarea  name="story" rows="8"  cols="45" >
    Enter your story here…
</textarea>
```

Creating Submit and Reset Buttons

- The submit and reset button input types let the user send form data to be processed or clear the form and start all over.
- The following code snippet shows addition of submit and reset buttons

```
<p>
<input type="submit" name="btn_submit" value="Submit Story" />
<input type="submit" name="btn_reset" value="Clear the form" />
</p>
```

Image can also be used for buttons as shown in the example below:

```
<input type="image" name="btn_image" src="submit.png"
   alt="Submit button" />
```


Creating Check Boxes

- Check boxes let users choose more than one option from a given set if they so desire.
- Check boxes are the way to go if users are to provide more than one answer to MCQs.

```
<p>Choose day(s) that you will be available:</p>
<p>
<input type="checkbox" name="day1" value="Sunday" />Sunday<br />
<input type="checkbox" name="day2" value="Monday" />Monday<br />
<input type="checkbox" name="day3" value="Tuesday" />Tuesday<br />
<input type="checkbox" name="day4" value="Wednesday" />Wednesday<br />
<input type="checkbox" name="day5" value="Thursday" />Thursday<br />
<input type="checkbox" name="day6" value="Friday" />Friday<br />
<input type="checkbox" name="day7" value="Saturday" />Saturday<br />
</p>
```

Creating Radio Buttons

- Radio buttons let users choose only one option from a given set of options.
- Radio buttons are the way to go if users are to provide only one answer to MCQs.

```
<p>Choose a day that we can contact you:</p>
<p>
<input type="radio" name="day" value="Sunday" />Sunday<br />
<input type="radio" name="day" value="Monday" />Monday<br />
<input type="radio" name="day" value="Tuesday" />Tuesday<br />
<input type="radio" name="day" value="Wednesday" />Wednesday<br />
<input type="radio" name="day" value="Thursday" />Thursday<br />
<input type="radio" name="day" value="Friday" />Friday<br />
<input type="radio" name="day" value="Saturday" />Saturday<br />
</p>
```


The *action* Attribute of the <form> tag

- When a user clicks a Submit button on a form, the content of the form is sent to the server.
- The form's ***action*** attribute defines the name of the file to send the contents to for further processing.
- The file defined in the action attribute usually does something with the received input.

HTML Form Example

```
<html>
    <head> <title> HTML Forms: </title> </head>
    <body>
    <fieldset>
        <legend> User Details: </legend>
         <form  action="process.php" method="post">
Your Name here:
         <input type="text" name="name"size ="30" /> <br />
Your Email here:
         <input type="text" name="email" size="30" /> <br />
Your Password here:
         <input  type = "password"   size = "20" /> <br />
Please browse and upload your comments file:
         <input  type="file"  size="45"> <br />
         <input type="submit" value="Send">
         </form>
         </fieldset>
    </body>
</html>
```


Introduction to Scripting

- Scripting is the act of writing little computer programs that can enhance the appearance and functionality of web page.
- Scripts are executed either at the server-side or at the client-side.
- This leads to two categories of scripting:
  - **Server-side scripting and**
  - **Client-side scripting.**

Server-Side Scripting - 1

- **Server-side scripting** is a web server technology in which a user's request is fulfilled by running a script directly on the web server to generate dynamic web pages.
- This is different from client-side scripting where scripts are run by the viewing web browser.
- It is usually used to provide interactive web sites that interface to databases or other data stores.
- The primary advantage to server-side scripting is the ability to highly customize the response based on the user's requirements, access rights, or queries into data stores.

Server-Side Scripting - 2

- Some commonly used server-side scripting languages are:
  - PHP (*.php)
  - ASP/ASP.NET (*.asp/*.aspx)
  - Perl (*.pl)
  - ColdFusion Markup Language (*.cfm)
  - Java via JSP(*.jsp)
  - Python (*.py)
  - Ruby (*.rb)

Client-Side Scripting

- Client-side scripting is a type of scripting that does all of its processing on the user's own computer.
- It is commonly used to create pop-up windows, instant-redirect pulldown menus, shopping-cart calculations, and mouseover effects (i.e., menus or images that change when the user's mouse passes over them).
- Javascript is the most popular client-side scripting language

The <script> </script> Tag Pair - 1

- The **<script > </script>** container is used to enclose the actual elements of the scripting language.
- The common attributes of this tag are:

  src, type and ***language***
- The ***src*** attribute is used to specify the URL of an external script into the HTML document.
- The external source file should contain no HTML codes and must use the appropriate file extension.
- **e.g**: ***<script src="coolscripts.js"></script>***

The <script> </script> Tag Pair - 2

- The type attribute should always be used with **<script>** container.
- It is used to specify the scripting language used for a particular script.

E.g

<script type="text/javascript">

alert("Javascripts codes go here!");

</script>

Scripts in Head Section

- Scripts to be executed when they are called, or when an event is triggered, go in the head section.
- When you place a script in the head section, you will ensure that the script is loaded before anyone uses it.

```
<html>
<head>
    <script type="text/javascript">.
    alert("Just saying Hi with JS in the head section!");
    </script>
</head>
</html>
```

Scripts in Body Section

- Scripts to be executed when the page loads go in the body section.
- When you place a script in the body section it generates the content of the page.

```
<html>
<head>
</head>
<body>
    <script type="text/javascript">
        alert("Just saying Hi with JS in the body section!");
    </script>
</body>
</html>
```


Scripts in Head and Body Sections

- You can place an unlimited number of scripts in your document, so you can have scripts in both the body and the head section.

```
<html>
<head>
    <script type="text/javascript">
        Javascript Statements
    </script>
</head>
<body>
    <script type="text/javascript">
        Javascript Statements
    </script>
</body>
</html>
```

Using External Scripts

- Sometimes you might want to run the same JavaScript on several pages, without having to write the same script on every page.
- To simplify this, you write a JavaScript in an external file.
- Save the external JavaScript file with a .js file extension.
- **Note:** The external script must not contain the **<script>** tag.
- To use the external script, point to the .js file in the "**src**" attribute of the **<script>** tag:

```
<script type="text/javascript" src="externaljsfile.js"></script>
```


Summary - 1

- Web developers use HTML forms to accept inputs from end users of web sites and web applications.
- The **<form> </form>** tag pair is used to surround the form controls; which are the real elements that allow end users to provide their inputs.
- The **action** and **method** attributes of the **<form>** tag are very important to add interactivity to web sites and web applications.

Summary - 2

- Scripting is lightweight programming.
- Scripting brings programmability to markup languages.
- Server-side scripts run on the server and send their result in a form of HTML codes to the browser for onward display to end users.
- Client-side scripts are interpreted by the scripting parsers of the browser architecture.
- CSS is the preferred technology for designing the presentation layer of websites and web applications

Take Home Activity -- 1

1. Create three related web pages and link them together using the **<a> </a>** tag.
2. Using the **<img />** tag, insert your own photograph in one of the pages created above.
3. Using the <audio> and <video> tags insert audio and video contents into the web pages you created.
4. Look out for any report from your personal records or place of work and present it as a table on your web page.

Take Home Activity -- 2

1. I have shared a sample of the EC's Voters Registration Form with your reps. Study the form carefully and go through the next stages of this activity.
2. Using the skills gained in this lecture, design the electronic version of the form.
3. Pay attention to the various information requirements on the form and use the most appropriate form controls in your design.

Our Next Lecture …

1. **Introduction to CSS**
2. **CSS Document Structure.**
3. **Applying CSS to Web Pages.**